

# Measure It, Graph It

## Answer Key

Grade 1

### Quick Answers

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- |            |                                         |
|------------|-----------------------------------------|
| 1. 5 cubes | 6. 20                                   |
| 2. $<$     | 7. 5                                    |
| 3. 5, 7, 8 | 8. 6 cubes                              |
| 4. 4       | 9. 2                                    |
| 5. Kim     | 10. (a) stickers Kim still needs = 3, — |
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### What is verified

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9 of 10 problems fully machine-checked against SymPy.  
 — marks an answer only you can judge — problem 10. The worked solution states what a correct response must contain.

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### Curriculum

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**Level:** Grade 1

1.MD – *problems 1, 2, 3, 4, 5, 6, 7, 8, 9, 10*

*Difficulty 1–4 of 5 across 11 tagged checks.*

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### Common wrong answers

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7. *If they got 13: added the two counts instead of finding how many more*

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1. Rope 9 cubes, scarf 4 cubes. How many cubes longer is the rope?

**Solution:** Same unit on both, so subtract the counts:  $9 - 4 = 5$ . The rope is longer by 5 cubes

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2. Worm 6 clips, leaf 8 clips. Write  $>$ ,  $<$ , or  $=$ :  $6 \_ 8$ .

**Solution:** 6 clips is fewer than 8 clips, so the worm is shorter:

$6 < 8$

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3. Pencil 7 cubes, marker 5 cubes, brush 8 cubes — order shortest to longest.

**Solution:** The smallest count is 5, then 7, then 8:

5, 7, 8

4. How many stickers does Ben have?

**Solution:** Count the dots in Ben's row of the chart: four dots.

4

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5. Who has the most stickers?

**Solution:** Compare the rows: Kim's row has the most dots (more than Ana's and Ben's).

Kim

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6. How many stickers do the three kids have all together?

**Solution:** Add the three rows:  $7 + 4 = 11$ , and  $11 + 9 = 20$ .

20

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7. How many more stickers does Kim have than Ben?

**Solution:** "How many more" means find the difference:  $9 - 4 = 5$ .

5

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8. Teacher's ribbon 14 cubes, Jon's ribbon 8 cubes. How many cubes shorter is Jon's?

**Solution:** Subtract the counts:  $14 - 8 = 6$ . Jon's ribbon is shorter by

6 cubes

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9. How many fewer stickers does Ana have than Kim?

**Solution:** "How many fewer" is also a difference:  $9 - 7 = 2$ .

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10. Challenge: Kim wants 12 stickers. (a) How many more does she need? (b) Say how the chart helps.

**Solution:** (a) The chart shows Kim has 9. Count on from 9 to 12: 10, 11, 12 — three jumps.

3

(b) *Instructor judges:* accept any wording that uses the chart as evidence — for example, "the chart shows Kim already has 9, so I counted on from 9 up to 12 and needed 3 more."